Tristan Hoellinger

Postdoctoral availability: October 2026.

Institut d'Astrophysique de Paris
tristan.hoellinger@iap.fr
hoellin.github.io
0000-0003-0217-8542

Academic Employment

October 2023 - September 2026 **PhD Candidate**, *Institut d'Astrophysique de Paris (CNRS – Sorbonne Université)*, Paris, France
Topic: Robust cosmological physics with mis-modelled galaxy clustering probes.
International collaborations (full member): Aquila Consortium (2023-), Euclid consortium (2023-)
Advisors: Florent Leclercq (CR CNRS/INSU & INS2I) and Guilhem Lavaux (DR CNRS INSU; HDR).

Research Interests

Numerical & Statistical Cosmology,
Implicit Inference, Field-level Inference, Robust Machine Learning,
Deterministic & Stochastic Optimisation, Bayesian Optimisation.

Languages

English Fluent — Cambridge Linguaskill Business: C1+ (highest score) French Native

Academic Service

2023 - 2024 **Co-organiser of the weekly Friday's pre-colloquium meetings at the IAP**
Weekly informal meeting between PhD students and invited researchers at the IAP.

2024 **Co-animated a workshop at the Fête de la Science 2024**, *Oct 13th, 2024*
Co-animated a workshop for 11+ years old children, "History of the Universe and CMB".

2023 **Member of the LOC for the Aquila 2024 autumn meeting**, *Oct 21st - Oct 25th, 2024*
Bi-annual meeting of the Aquila consortium, IAP, Paris.

2023 **Member of the IAP LOC for the ML-IAP/CCA conference**, *Nov 27th - Dec 1st, 2023*
Conference on Machine Learning in astronomical surveys, IAP, Paris / Flatiron institute, New York.

Teaching

2024 - 2025 **Teaching assistant in Mathematics, 76 hours**, *3rd yr BSc. in Mathematics*, Sorbonne University
Differential manifolds, Stokes formulae & PDEs (38 hours), (un)constrained optimisation (38 hours).

2023 - 2024 **Teaching assistant in Mathematics, 64 hours**, *3rd yr BSc. in Engineering*, Sorbonne University
Distribution theory & Fourier analysis (16 hours), Fourier analysis & distribution theory (16 hours), ODEs (second year bachelor, 18 hours), numerical methods (14 hours).

Education

2023 - 2026 **PhD in Astronomy & Astrophysics**, *Sorbonne Université*, Institut d'Astrophysique de Paris

2020 - 2022 **Dual MSc in Applied Mathematics**, *INSA Toulouse & N7*, Toulouse, France
MSc in Mathematics (First-Class) and MSc in Artificial Intelligence & Computer Sciences (First-Class).

2019 - 2020 **Academic Exchange in Fundamental Physics**, *Paris-Saclay University*

2016 - 2019 **BSc in Mathematics and Computer Sciences**, *INSA Toulouse*

Publications

2025 Diagnosing systematic effects using the inferred initial power spectrum, Hoellinger and Leclercq, A&A 699, A224 (2025), DOI: 10.1051/0004-6361/202453416, arXiv:2412.04443.

2025 A Vorticity Confinement correction for discontinuous Galerkin schemes applied to fluid flow problems, Hoellinger, Manueco and Chapelier, 2025, IJNMHFF, 10.1108/HFF-11-2024-0854.

2023 Enhancer/gene relationships: need for more reliable genome-wide reference sets, Hoellinger et al., 2023, Front. Bioinform., Insights in Integrative Bioinformatics, PMC9999192

2020 Data-driven simulation for augmented surgery, Mendizabal et al. (incl. Hoellinger), 2020, Developments and Novel Approaches in Biomechanics and Metamaterials. Springer, hal-02538101

Invited Talks

September 23rd 2025 **Astrocoffee — Journal club**, *Observatoire astronomique*, Strasbourg, France
Accurate Small Scale Dynamics in COLA, *20 minutes*.

June 27th 2025 **Cosmology & Gravitation Seminar**, *Oskar Klein Centre*, Stockholm, Sweden
Robustifying cosmological inference against model misspecification, *30 minutes*

March 6th 2025 **Cosmology Group Meeting**, *ICUUB — Institute of Cosmos Science*, Barcelona, Catalunya, Spain
Diagnosing systematic effects in field-based, implicit likelihood cosmological inference, *40 minutes*.

Contributed Talks at International Conferences

July 20th **École de Physique des Houches: The Dark Universe**, *Les Houches*
Accurate Small Scale Dynamics in COLA, *20 minutes*.

Oct 17th 2024 **X Meeting on Fundamental Cosmology**, *Universidad de Sevilla*, Sevilla, Spain
Lightening blackbox models in field-based cosmological inferences, *15 minutes*.

Jan 25th 2024 **Euclid France 12th Symposium**, *IP2I*, Lyon
Implicit likelihood inference in cosmology while checking for survey systematics, *12 minutes*.

Nov 6th 2023 **Action Dark Energy Colloquium 2023**, *LAPP*, Annecy
Implicit likelihood inference in cosmology while checking for survey systematics, *15 minutes*.

Other Talks

May 23rd 2025 **PhD day**, Sorbonne Université, campus Pierre-et-Marie-Curie, Paris, France
Diagnosing Systematic Effects Using the Inferred Initial Power Spectrum, *5 minutes*.

Apr 10th 2025 **The Elbereth Conference 2025**, *by & for PhD students*, Observatoire de Paris, Meudon
Diagnosing Systematic Effects Using the Inferred Initial Power Spectrum, *10 minutes*.

August 28th 2024 **Rodolphe Clédassou Summer School**, *by & for PhD students*, Azureva, Hendaye
Lightening blackbox models to deal with systematic effects, *5 minutes*.

June 7th 2024 **PhD students Seminar**, *IAP*, Paris, France
Lightening blackbox models, *40 minutes*.

March 29th 2024 **PhD day**, Institut d'Astrophysique de Paris
Novel methods for implicit likelihood inference in cosmology, *10 minutes*.

Feb 28th 2024 **The Elbereth Conference 2024**, *by & for PhD students*, IAP, Paris
Implicit likelihood inference in cosmology while checking for survey systematics, *12 minutes*.

Leadership and Community engagement

2025 - 2026 **Member of the Committee against harassment and discriminations at the IAP**, *Paris*

2019 **Mentor of high-school students within the Equal Opportunity Program Ô Talents**, Toulouse

2019 **President of the Theatre troupe "La Catin"**, *6 months*, Toulouse

International conferences, schools and workshops

2023 - 2026 **Aquila Consortium Workshops**, Paris, Oxford, Stockholm, 1 week twice a year

2025 **École de physique des Houches: The Dark Universe**, Les Houches, 1 month

2024 **Rodolphe Clédassou School on Cosmology and large-scale surveys**, *Hendaye*, 2 weeks

2024 **MaxEnt 2024**, *Faculty of Economics, Ghent University*, Ghent, Belgium, 1 week
43rd International Workshop on Bayesian Inference and Maximum Entropy Methods.

2023 **Future Cosmology**, *Institut d'Études Scientifiques de Cargèse*, Corsica, 1 week

2023 **Euclid School on Cosmology and large-scale surveys**, *Ronces-les-Bains*, 2 weeks

Grants

2023 PhD student fellowship "Politique Scientifique" from Sorbonne Université. Value ∼100K€. This program aims to support particularly promising thesis projects, backed by high-quality applications. Only one fellowship awarded for Astronomy & Astrophysics in 2023 (8 in total).

2016 - 2020 National scholarship + merit-based grant holder during my undergrad studies (∼ 7k€/year).

Traineeships

March 2023 - July 2023 **Graduate Trainee**, *Institut d'Astrophysique de Paris*, Paris
Trainee in the "Large-scale structure and distant Universe" group at the IAP.
Advisors: Florent Leclercq & Guilhem Lavaux

February 2022 - February 2023 **Master thesis**, *ONERA - The French Aerospace Lab*, Châtillon
Research trainee in the NFLU and MSAT teams. I developed a Vorticity Confinement correction for the direct numerical simulation of Navier-Stokes equations through Discontinuous Galerkin Methods, at arbitrarily high order. I implemented the method in a massively parallel solver of unsteady compressible turbulent flows. Advisors: Jean-Baptiste Chapelier & Lucas Manueco.

September 2020 - January 2022 **Junior Bioinformatics Scientist**, *Inserm*, Toulouse
In parallel to my Master's degree in Mathematics, I worked as a junior researcher in quantitative epigenomics. I studied statistical methods to predict enhancer-gene interactions in human genomes based on high-throughput data, towards application to the search of variants involved in haemochromatosis or complex genetic disorders. Advisor: Sarah Djebali.

July 2021 - August 2021 **Research Trainee — Visiting**, *Centre for Genomic Regulation*, Barcelona
Guigó lab. I created a Nextflow pipeline based on surrogate variable analysis for automated identification of genetic variants associated with intron retention estimated from RNA-sequencing in blood cell types from the Blueprint Project. Advisor: Diego Garrido.

July 2020 **Research Trainee**, *Toulouse Mathematics Institute*, Toulouse
Stochastic calculus: theoretical study of the convergence rate of quantum non-demolition measurements, under the supervision of Tristan Benoist & Clément Pellegrini.

April 2020 - June 2020 **Research Trainee**, *IRAP - Institut de recherche en astrophysique et planétologie*, Toulouse
Theoretical study of kinetic scale plasma turbulence in the solar wind. Advisor: Philippe Louarn.

June 2019 - August 2019 **Research Trainee**, *Inria, National Institute for Research in Digital Science and Technology*
I analysed the robustness of a deep learning method for real-time biomechanical simulation, with respect to data sparsity and noise, in the context of intraoperative augmented surgery. Advisor: Andréa Mendizabal.

July 2017 **Farm Hand**, *La Ferme aux 100 Blés*, Saint-Broing-les-Moines
Agricultural operation. I used various tools (pneumatic hammer, freight elevators, riveting guns), packed flour and oil, made sourdough bread, ran a market stall and delivered farm products to clients.

Education — Details

2023 - 2026 **Ph.D. Candidate**, *Sorbonne Université (Institut d'Astrophysique de Paris)*, Paris, France
Title: Robust cosmological physics with mis-modelled galaxy clustering probes.

2020 - 2022 **M.Sc. in Mathematics & M.Sc. in Hybrid AI**, *INSA Toulouse x N7*, Toulouse
This is a transversal curriculum leading to 2 *Diplômes d'ingénieur* (2 distinct MS degrees from French leading *grandes écoles*), built around 3 main axes: numerical mathematical modelling, statistical mathematical modelling, hybrid artificial intelligence. **Humanities are part of the curriculum**. Major scientific topics:

- Deterministic and stochastic optimisation, variational data assimilation, optimal control theory,
- PDEs, finite volume and Galerkin methods, large linear systems, signal / image / wavelets,
- Advanced statistics, reinforcement learning, physically constrained machine learning.

2019 - 2020 **Academic exchange in Physics**, *Paris-Saclay University*, Gif-sur-Yvette
I took a full year off with respect to my education in Mathematics at INSA Toulouse, in order to develop my knowledge of fundamental physics. I was enrolled in the *Magistère de Physique d'Orsay*, where I studied for one successful semester (hamiltonian mechanics, quantum mechanics, general relativity and cosmology, statistical physics). I completed my academic year with 2 traineeships in theoretical physics.

2016 - 2019 **B.Sc. in Mathematics and basic sciences**, *INSA Toulouse*, Toulouse
First part of a selective public 5-year curriculum leading to a *Diplôme d'ingénieur (M.Sc.) in Mathematics*. Major: mathematics. Other courses: computer sciences, fundamental and applied physics, engineering.

2016 - 2017 **Propaedeutic year**, *INSA Strasbourg*, Strasbourg, *Top 10%. Transfer to INSA Toulouse.*

2013 - 2016 **High school diploma in sciences**, *Lycée Charles-Emile Freppel*, Obernai, France
Public high school. Highest Honours — Top 2% countrywide.

Practical skills in Applied Maths & in Computer Science

Generalities Signal/image processing, data assimilation, optimisation (smooth, convex, stochastic), inverse problems.

Numerical simulation Fluid dynamics, finite volume methods, Galerkin methods, particle methods.

Statistics Tests, stochastic processes, sampling methods, machine learning: kernel methods, neural networks (CNNs, RNNs, PINNs), variational data-assimilation networks, wavelet scattering transforms.

HPC Solving large, ill-conditioned linear systems, multiprocessing, parallel computing (OpenMP, MPI).

Programming Python, C/C++, Fortran, Julia, R.

Misc Git, SLURM, Docker, AWS.

Miscellaneous

Hobbies Portrait photography, sport (running, bouldering, hiking).

Referees

Upon request.